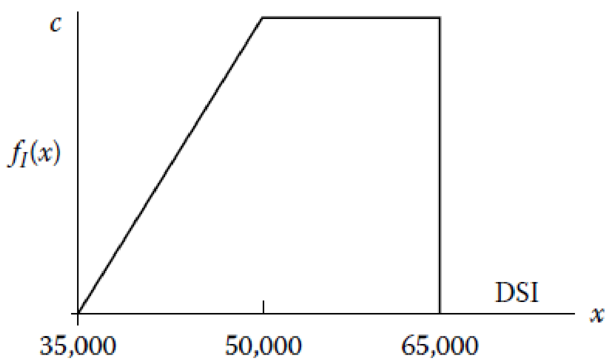
EMSE 6420 – Spring 2025
Graded Problem Set #3 (**40 points**)

Directions: Please note that you have to work individually on this problem set. You are not allowed to discuss this homework assignment with any student. Please show all your work. The problem set must be submitted via the Blackboard Learn platform before the **due time (11:59 pm) on Thursday, February 27th, 2025.**
Hint: For this homework assignment, consider referring to the standard formulas provided for specific distributions in Chapter #4 of the uploaded textbook.

Problem #1 – Expectation value and variance of continuous random variables. Suppose the uncertainty in the size ***I*** of a software application (DSI) is expressed by the PDF in the following figure, where the constant c equals $\frac{1}{22500}$.



The PDF and CDF of this continuous random variable are presented as below, respectively:

$$f(x) = \begin{cases} \frac{1}{337500000} \cdot (x - 35000) & if\ 35000 \le x < 50000 \\ \frac{1}{22500} & if\ 50000 \le x < 65000 \\ 0 & otherwise \end{cases}$$

$$F(x) = \begin{cases} 0 & if\ x < 35000 \\ \frac{1}{675000000}(x - 35000)^2 & if\ 35000 \le x < 50000 \\ \frac{1}{3} + \frac{1}{22500}(x - 50000) & if\ 50000 \le x < 65000 \\ 1 & if\ 65000 \le x \end{cases}$$

(1). Please find the expectation value $\boldsymbol{E}(\boldsymbol{x})$ of this continuous random variable. **(3 points)**

(2). Please find the **<u>standard derivation</u>** σ of this continuous random variable. **(3 points)**

(3). Please compute the probability: $\boldsymbol{P}(|x - E(x)| > \sigma)$**. (2 points)**

Problem #2 – Calculation of expectation value and variance of linear combination of random variables. Suppose X is a continuous random variable. Its expectation value is $E(X) = 4$, and its variance is given by $Var(X) = \frac{E(X)}{2}$. Now, Y is another continuous random variable but it presents a linear transformation form of X, given by $Y = \frac{1-2X}{2}$.

(1). Please compute the expectation of the random variable Y. **(2 points)**

(2). Please compute the variance of the random variable Y. **(2 points)**

Problem #3 – Moment of random variable. Please derive a general formula for the k-th moment of a continuous random variable X with the probability density function $f_X(x) = \frac{1}{b-a}$,where a and b are constant and $a \leq x \leq b$. **(3 points)**

Problem #4 – Special distributions (Beta Distribution). Suppose the continuous random variable $Y \sim Beta(\alpha, \beta)$ and its PDF is given by $f_Y(y) = 12 \cdot y^2 \cdot (1-y)$, where $0 < y < 1$. *Hint: In this case, the random variable Y follows the standard Beta Distribution.*

(1). Please find the value of constant α and β. **(2 points)**

(2). Please compute the value of $E(Y) + \sigma_Y$, where $E(Y)$ is the expectation value and σ_Y is the **standard derivation**. **(3 points)**

(3). Please determine the probability $\boldsymbol{P}(0.3 < Y \leq 0.7)$. **(2 points)**

Problem #5 – Special distributions (Log-Normal Distribution). Suppose the uncertainty in a system's cost is described by a *lognormal* PDF with $E(Cost) = 25$ ($\$M$) and $Var(Cost) = 225$ $(\$M)^2$.

(1) Please compute the probability $P(Cost > E(Cost))$. **(3 points)**
(2) Please compute the probability $P(Cost \leq 50)$. **(2 points)**

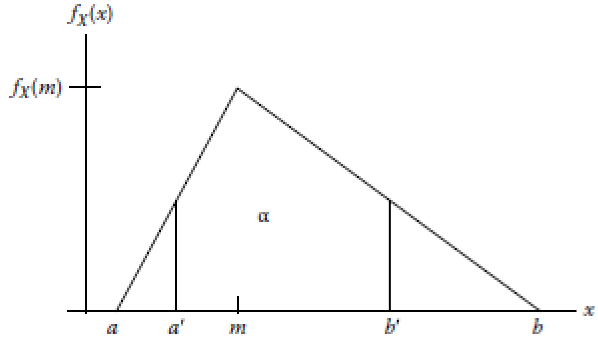
Problem #6 – Specify the Special distributions (Beta Distribution). Suppose I represents the uncertainty in the number of DSI for a new application. Suppose a team of software engineers judged 35000 DSI as a reasonable assessment of the 50th percentile of I and a size of 60,000 DSI as a reasonable assessment of the 95th percentile. Furthermore, suppose the uncertainty in the number of DSI is subject to beta distributions, given by $\boldsymbol{Beta}(\mathbf{2}, \ \frac{\mathbf{7}}{\mathbf{2}}, \ \boldsymbol{a}, \ \boldsymbol{b})$

(1). Find the extreme possible values for *I*, that is to say, what is the value of constant ***a*** and ***b***? **(4 points).**
For simplification purposes, you are provided with the estimated point values derived from the corresponding Standard Beta Distribution $Y \sim Beta(2, \ \frac{7}{2})$ *where* $y_{0.5} = 0.34086$, *and* $y_{0.95} = 0.70189$.

(2). Compute the mode of *I*. **(2 points)**

(3). Compute the expectation value, $E(I)$, and the standard derivation, σ_I. **(2 points)**

<u>Problem #7 – Specify the Special distributions (Triangular Distribution)</u>. Suppose I represents the uncertainty in the amount of new code for a software application. Suppose this uncertainty is characterized by the triangular PDF shown as below.



Assume for this case, $\frac{P(X \le a')}{P(X \ge b')} = \frac{P(X \le m)}{P(X \ge m)}$, which secures the eligible conditions of using the conclusions of specifying a triangular distribution.

If the probability is 0.90 that the amount of code is between 20,000 and 30,000 DSI, with 25,000 DSI as most probable, determine the expectation value $E(I)$. **(5 points)**